# UNIT – V

**Man–Machine Interaction • Displays • Controls • Design Guidelines • Standards**

**Syllabus (as given)**

Man–Machine Interaction: man–machine interaction cycle, man–machine interfaces, displays – factors that control choice of display, visual displays – qualitative displays: moving pointer displays, moving scale displays, digital displays, indicators, auditory displays, tactile displays. Factors affecting effectiveness of displays. Quantitative displays, check-reading displays, representational displays. Types of controls and their integration with displays. Design guidelines for displays and controls: viewing distance, illumination, angle of view, reach etc.; general design checklist for displays and controls. Standards for ergonomics in engineering and design, displays and controls.

## 1. Man–Machine System and the Interaction Cycle

- A **man–machine system** (human–machine system) is a combination of one or more humans and one or more machines (equipment, tools, computers, vehicles) which interact to perform a function in a given environment. Examples: driver–car, operator–lathe, pilot–aircraft, user–smartphone.
- Types by degree of human control: **Manual** (human supplies power & control – hand tool), **Mechanical / semi-automatic** (machine gives power, human controls – lathe, car), **Automatic** (machine does sensing, decision & action; human monitors & maintains – CNC, robot, process plant).
- Why study? Most accidents & errors are due to poor interface design; system performance = performance of the human + machine + the way they are linked.

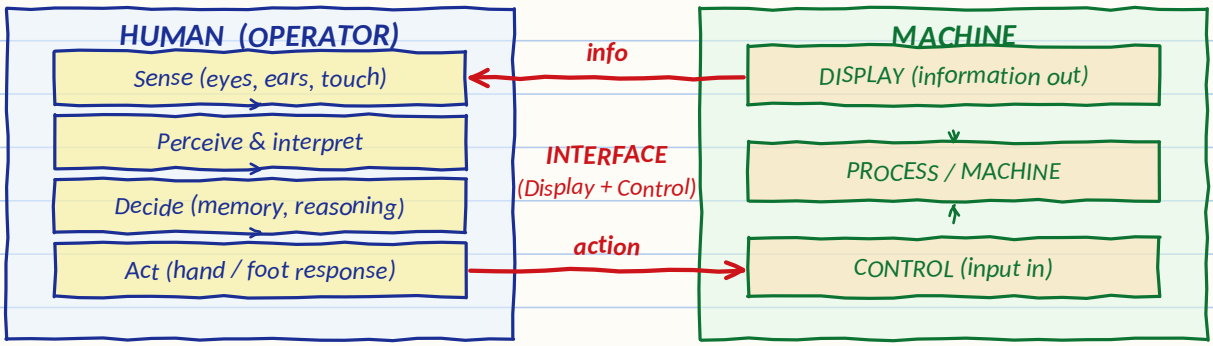



*Fig 5.1 Man–machine interaction cycle*

- **Interaction cycle:** Machine state shown on **display** → operator **senses** the signal → **perceives** & interprets using memory → **decides** what to do → **acts** on the **control** → machine changes state (process) → new output on display (feedback) → the cycle repeats. It is a **closed-loop** system; time delay / lag in any part reduces performance.
- **Man–machine interface:** the point(s) where information / energy is exchanged – **displays** (machine → man), **controls** (man → machine), plus the work-place layout, environment, software / GUI, manuals, communications.

| Humans are better at | Machines are better at |
|---|---|
| Perceiving patterns, judging, improvising, flexibility | Fast, repetitive, precise work; high force & power |
| Handling unexpected events, inductive reasoning | Storing & recalling huge data accurately; deductive logic |
| Sensitive detection of small signals; selecting | Monitoring for long periods without fatigue (vigilance) |
| Creativity, ethics, value decisions | Operating in hazardous environment; simultaneous tasks |

Note: Function allocation ("MABA-MABA", Fitts list): assign each function to the one – human or machine – that does it better; keep the human in control (supervisory role).

## 2. Displays – Functions, Classification and Choice

- **Display** = a device that presents information about the machine / process to the operator (by sight, sound or touch). Functions: give **status** information, **quantity,** warn of danger, instruct, show trend, help in control (tracking), and identify.

| Classification basis | Types |
|---|---|
| Sense used | **Visual, auditory, tactile** (haptic) displays |
| Nature of information | **Quantitative** (numerical value), **qualitative** (status / trend / direction), **check-reading** (normal or not), **representational** (pictorial / symbolic), alphanumeric, warning / indicator |
| Time variation | Static (signs, charts, labels) and dynamic (dial, CRT, LED, LCD) |
| Technology | Mechanical (analogue dial), electronic (digital, LED, LCD), screens, projection, HUD, VR / AR |

## Factors that control the choice of display

1. **Type of information** to be shown (exact value → digital / quantitative; trend / rate → analogue; status → light; warning → sound).
2. **Accuracy and speed** of reading required; **resolution:** response time.
3. Task of the operator: reading, tracking, check reading, setting, monitoring; need for rate-of-change and trend.
4. **Environment:** illumination level, noise, vibration, dust, temperature, glare, darkness.
5. **Space available** on the panel; distance & angle of viewing; number of displays; operator position and mobility.
6. Operator characteristics: visual acuity, training, age, expectations (stereotypes); other loads on the sense (visual overload → use auditory).
7. Cost, reliability, maintainability; compatibility with controls and with other displays.

| Use VISUAL display when | Use AUDITORY display when | Use TACTILE display when |
|---|---|---|
| Message is complex, long, or needs later reference; relates to location in space; auditory channel overloaded; environment noisy; person stays at one place | Message is simple, short & urgent (warning); calls for immediate action; person moves about; visual channel overloaded; dark / poor lighting; no need for later reference | Both visual & auditory channels are busy or unusable (dark, noise); for identification of controls by feel; blind users (Braille); alerts (vibration) |

## 3. Visual Displays

### (a) Qualitative displays

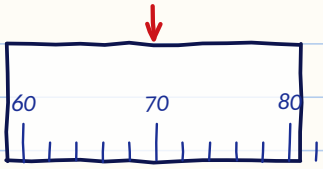
- Give **approximate / non-numerical information:** status (on–off, safe–danger), condition (hot–normal–cold), trend, rate of change or direction of change, rough value. Use coloured zones or simple symbols on the dial (e.g. green = normal, yellow = caution, red = danger).
- Examples: a fuel gauge (E – ½ – F), engine-temperature gauge with a green band, tyre-pressure indicator with limits, signal lamp.
- Advantage: fast reading, a glance shows whether things are normal; less chance of error; takes less space.

## (b) Quantitative displays – dynamic types

- Give **exact numerical value** of a variable (speed, temperature, pressure, time). Three main types:




(1) Moving pointer, fixed scale




(2) Moving scale, fixed pointer (window)
(scale slides behind pointer)




(3) Digital display (counter / LED / LCD)

Fig 5.2 Types of quantitative visual displays

| Characteristic | Moving pointer – fixed scale | Moving scale – fixed pointer | Digital (counter) |
|---|---|---|---|
| Quantitative reading | Fair (needs scale reading) | Good if scale not too long; poor if moving fast | Best – fastest & most accurate for exact value (fewest errors) |
| Qualitative / trend / rate-of-change | Good – movement of pointer gives rate & direction | Poor – the moving scale is hard to follow | Poor – numbers jump; no rate cue |
| Setting (matching a value) | Good – control–pointer relation clear | Fair | Good if counter turned directly by knob |
| Tracking | Good – best | Poor | Poor |
| Space needed | Large (long scale) | Small (only window shown) – long range in little space | Very small |
| Check reading | Good – pointer position gives quick cue | Poor | Poor |

## Design of scales & pointers (quantitative visual displays)

- **Scale length** matched to the required accuracy (more divisions for more precise reading); open (partial) circular or horizontal scales are preferred; vertical scales read in the same direction as the change (up = increase).
- Graduations: **major marks numbered, with minor marks between;** number progression by 1, 2, 5, 10 or multiples (avoid 3, 4, 6, 7, 8 as scale increments); minimum 1 : 2 or 1 : 5 sub-division; do not use more than 9–10 minor marks between numbers; avoid fractions.

- Numerals increase **clockwise** (circular), left → right (horizontal), bottom → top (vertical); numbers upright and **outside** the scale marks (not hidden by pointer); numerals in simple type face (sans-serif), black on white (or white on black) for contrast.
- **Pointer:** simple, tapered, tip touches the smallest graduation (no gap, no overlap); close to the dial face (to avoid parallax); same colour as marks; counter-balanced; length reaches the smallest mark.
- Colour coding of dial zones (green / yellow / red) for qualitative information; marks at the normal operating value.
- **Parallax** error: due to gap between pointer and scale – viewer's eye not perpendicular; reduce with small gap, mirrors on the scale.

### (c) Check-reading displays

- Used to find quickly whether a value is **normal or deviating** – e.g. in panels of many identical dials: all pointers are arranged in a line (e.g. at the "9 o'clock" or "12 o'clock" position); any deviation is easily seen by the change in pattern. Quantitative reading not needed.

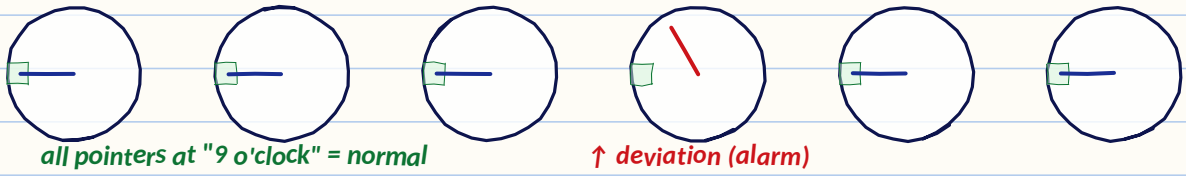



*Fig 5.3 Check-reading (comparison) display panel*

### (d) Representational (pictorial) displays

- Show the **relationship / situation as a picture or symbol** so that the operator understands the system "at a glance": aircraft attitude (artificial horizon) indicator, map / navigation display (GPS), **mimic diagrams** of a power plant or pipe network, HUD in cars / aircraft, CAD or SCADA graphical screens, radar displays.
- Principles: **pictorial realism** (looks like the real thing), **moving-part principle** (moving element moves like the real thing), **principle of the colocated** / proximity compatibility (related information close together), minimise clutter, de-clutter by layers.

### (e) Indicators / signal lights (qualitative status displays)

- Used for **status / warning / alarm** (on-off, ready, fault). Design: size (visual angle ≥ 1°), brightness greater than surroundings (not dazzling), position within central visual field, flashing for urgency (3–10 flashes / s; "on" time ≥ 0.05 s; flash rate 1–2 Hz for attention; flashing lights are intrusive – use only for urgent condition), and legends beside the lamp.

| Colour | Meaning (IEC 60073 / common practice) | Typical use |
|---|---|---|
| Red | Danger, emergency, stop, fault, abnormal – needs immediate action | Emergency stop button, over-temperature, alarm |
| Yellow / Amber | Caution, attention, abnormal condition developing, near limits | Pre-warning, low level |
| Green | Normal, safe, ready, "go" | Machine running, supply on |
| Blue | Mandatory action; advisory information | Reset / required operator action |
| White / grey | Neutral, general information; no specific meaning | Power on, auxiliary |

Note: Do not use more than ≈ 5 colours; avoid colour as the only cue (colour-blind ≈ 8 % of men) – back up with position, shape or label.

## 4. Auditory Displays

- Convey information by **sound:** warning, alarm, signal, speech message, feedback of key press. The ear can receive from all directions, is always "open", and responds faster (≈ 0.15 s) than the eye – good for urgent warnings.

| Type | Characteristics / use |
|---|---|
| Pure tones / beeps | Single frequency; feedback, "ready", simple alert; may be masked by noise |
| Complex tones, chimes, gongs | Several frequencies; pleasant, e.g. station announcements, appliance alerts |
| Buzzers & bells | Low-frequency harsh buzz – equipment failure; bells – telephone, fire, school; high intensity |
| Horns, sirens | Loud, rising–falling pitch – emergency / evacuation, vehicles; detectable at a distance |
| Speech (voice) messages | Give complex information (aircraft "pull up"; train announcement); needs good intelligibility & language |
| Auditory icons / earcons | Short, symbolic sounds representing events (computer, mobiles, cars) |

- Design guidelines:
  - Frequency in the range **200 – 5000 Hz,** best 500 – 3000 Hz (ear most sensitive); for long distance and through obstacles use < 1000 Hz.
  - Intensity at least **10 – 15 dB above the masked threshold** of the background noise at the operator's ear (but < 120 dB, avoid startle / hearing damage); allow volume control for non-critical sounds.

- Make the signal **distinct** from the background noise & from other signals (different frequency, interrupted / modulated; 0.2 – 0.5 s beeps, 1–8 / s); use **intermittent** rather than continuous tones; ensure the sound is meaningful & learnt; use maximum 4–5 different warning tones.
- Ensure signal is audible everywhere in the area (avoid reflection); do not use if the ambient noise is extremely high or the listener is wearing hearing protection; provide a silence / acknowledge switch.
- Use a combined visual + auditory warning for critical alarms (redundancy), and give priority levels (urgent = faster pulses, higher pitch).

## 5. Tactile (Haptic) Displays

- Information given through the **sense of touch** – skin receptors (pressure, vibration, temperature). The fingertips are most sensitive (two-point discrimination ≈ 2 mm).
- Uses: **shape / size / texture coding of controls** (so a pilot or driver can identify them without looking – flap lever has flap-shaped knob, landing-gear lever has wheel-shaped knob); **Braille** and raised graphics for blind users; vibration alerts (mobile phone, steering wheel, joystick – "haptic feedback"); pin-matrix devices; force-feedback in tele-operation / surgery robots & virtual reality.
- Advantages: not affected by darkness, noise or glare; private; leaves eyes / ears free. Limitations: low information rate; needs skin contact; affected by gloves, cold, vibration; limited coding (≈ 6–10 shapes).
- Guidelines: textures should be clearly different (smooth, knurled, fluted); shapes distinguishable with gloves; vibration frequency ≈ 150–300 Hz (most sensitive: ≈ 250 Hz); use **at least 3 mm height difference** / distinct size steps for size coding; consistent meaning.

## 6. Factors Affecting Effectiveness of Displays

| Factor | Effect / design remedy |
|---|---|
| Contrast & brightness | Low contrast reduces legibility; keep contrast ratio ≥ 3 : 1, preferably 5 – 10 : 1 between figure and ground; black on white / white on black |
| Illumination & glare | Too low → eye strain; too high / glare → reflections on glass cover; use matt finish, shield, correct level 200 – 500 lux |
| Size of characters / viewing distance | Visual angle ≥ 20 – 22 minutes of arc; increase for poor illumination, vibration, fast reading |

| Factor | Effect / design remedy |
|---|---|
| Angle of view & position | Within the normal visual field (30° cone); avoid parallax; normal to the line of sight |
| Colour & coding | Use standard / stereotype colours; avoid too many; maintain colour contrast; not only colour |
| Clutter / information load | Too many displays or signals hamper detection; group & prioritise; remove unnecessary data |
| Operator factors | Visual acuity, age, colour vision, training, expectation, fatigue, stress |
| Environment | Vibration, glare, ambient noise (for auditory), dust, temperature; operating speed / time of exposure |
| Compatibility | Display and control relation, population stereotypes, language, symbols, consistent layout |

## 7. Controls – Types, Coding and Integration with Displays

- **Control** = a device through which the operator transmits an **input / command** to the machine (hand, foot, voice, eye). Functions: **activate / turn on-off, discrete setting** (select among positions), **continuous setting / adjust, data entry, tracking / steering.**

| Type of control | Examples | Used for |
|---|---|---|
| Discrete – two position | Push button, toggle switch, rocker, key switch, foot switch | On / off, start / stop, emergency stop |
| Discrete – multi-position | Rotary selector, thumb-wheel, slide / detent switch, push-button keypad | Selection of mode / speed / channel (3 – 24 positions) |
| Continuous – rotary | Knob, hand-wheel, crank, steering wheel | Fine setting (volume), steering, large force (hand-wheel) |
| Continuous – linear | Lever (hand / joystick), slider, push-pull knob, pedal | Gross positioning, speed (throttle), brake, tracking |
| Data-entry | Keyboard, touch screen, mouse, voice input | Text, command, pointing |

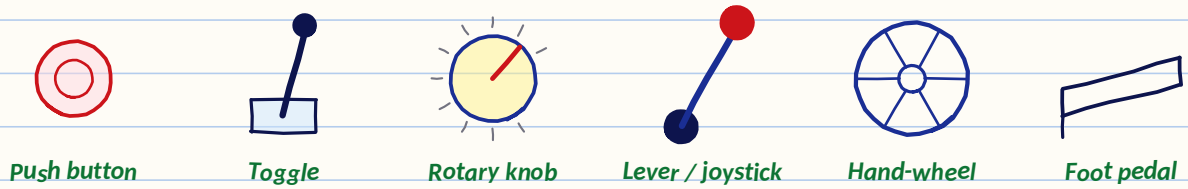



*Fig 5.4 Common types of controls (sketch)*

## Coding of controls (so that they are not confused)

- **Shape** (best – can be identified by touch), **size** (3 – 4 sizes), **colour** (red for emergency / stop), **location / position** (fixed place, standard layout), **texture / surface** (knurled, smooth), **mode of operation** (push, pull, turn), **labelling** (word or symbol – needs vision & light). Combine two or more codes for critical controls.
- **Resistance of controls:** elastic (spring) – gives feedback, good for fine setting; static friction (stiction); viscous damping (smooth, stable); inertia (flywheel, for steady motion). Resistance prevents accidental activation and gives "feel".
- Prevent accidental operation of critical controls: recessed or **guarded / covered,** interlock / lock, sequence of two actions, enough resistance, separation, orientation; emergency stop – large red mushroom head, easy to reach, push-to-lock / twist-to-release.

## Control–display relationships (integration)

- **Compatibility:** movement of control and response of display / machine should match user expectation – **spatial** (location), **movement** (direction), **conceptual** (symbol meaning), modality. Stereotypes: clockwise / up / right / forward = increase, on, start; counter-clockwise / down / left / back = decrease, off.
- **Warrick's principle:** the pointer of the display should move in the **same direction as the nearest point on the control** (for a knob next to a dial: clockwise turn moves the nearest part of the knob and the pointer in the same direction).
- **Control–display (C/D) ratio** = (movement of control) / (movement of display) (e.g. degrees of knob turn per cm of pointer travel). Low C/D → quick but coarse; high C/D → slow but fine. There is an **optimum C/D ratio** that minimises the total time = gross movement time + fine adjustment time (for knobs ≈ 0.2 – 0.8 depending on size / task).
- **Gain** = display change / control change (reverse of C/D); lag (time delay) and backlash should be minimised.

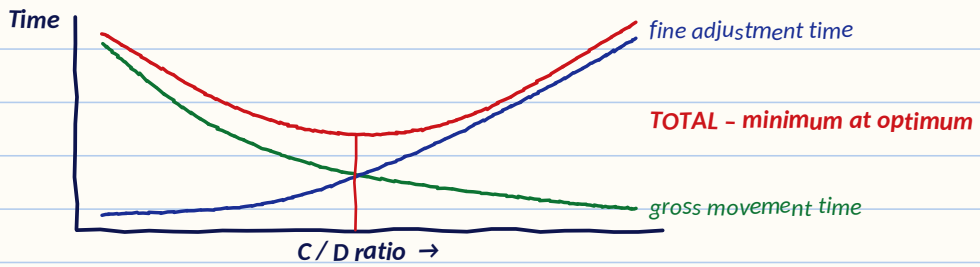



Fig 5.5 Optimum control–display ratio

### Integration / arrangement of displays and controls on a panel

- Arrange by **importance and frequency of use** (most important & frequent in the best position – central visual field and easy reach); by **function** (related items together, outlined / colour-grouped); by **sequence of use** (left → right, top → bottom); by **control–display link** (each control placed close to / below its display; use same layout, labels, colours; avoid hand covering the display – place control below or to the right of the display for right-handed users).
- Natural mapping: layout of controls mirrors layout of the controlled things (e.g. four stove burners vs four knobs – put knobs in same 2×2 pattern); consistency and standardisation (e.g. the same layout in all vehicles).

## 8. Design Guidelines for Displays and Controls

### (a) Viewing distance

- Normal viewing distance for dials / panels ≈ **50 – 70 cm** (not less than ≈ 30 cm – accommodation limit; VDT 50 – 70 cm). The size of characters must increase in proportion to distance.

**Character height from visual angle**

**$H = (D \times \theta) / 3438$** H = height of character, D = viewing distance (same units), θ = visual angle in minutes of arc (1 radian = 3438 minutes).

Recommended visual angle: 20 – 22 min (min. 16 min for good conditions); 30 – 45 min for poor illumination / important / fast-reading tasks.

Example: D = 700 mm, θ = 20 min → H = 700 × 20 / 3438 = **4.07 mm** (≈ 4 mm); for θ = 40 min H ≈ 8.1 mm.

### (b) Illumination

- General level 200 – 500 lux for reading dials and panels (higher, ≈ 500 – 1000 lux, for fine detail); uniform; no glare, reflection or shadows on the face of the display – matt glass, hoods, correct position of luminaires.
- For night / dark-room control (aircraft, ships, control rooms): **red or low-level white light** to keep dark adaptation; individually adjustable (dimmer) back-lighting; luminance of self-lit display not > 3 times the surrounding.

### (c) Angle of view

- The normal line of sight is about **15° below the horizontal** (in standing) and **≈ 30° – 35° below horizontal** for seated, relaxed head / eye position; displays should lie within **30° cone** around it (critical ones within ≈ 15°); maximum 60° with head movement.

- Horizontal: optimum ± 15° from the median plane; max. eye rotation ± 35°; head turn up to ± 95° (avoid). Mount displays **perpendicular to line of sight** (± 15° tilt) to avoid parallax and reflection.

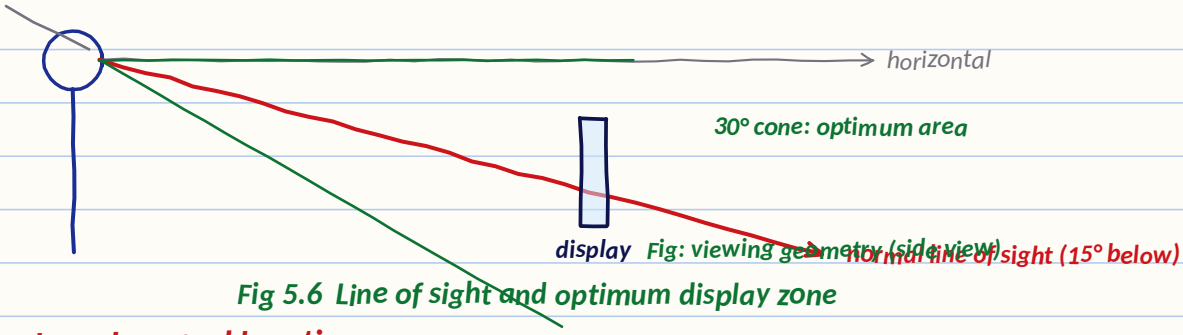



Fig: viewing geometry (side view)

Fig 5.6 Line of sight and optimum display zone

## (d) Reach and control location

- Locate frequently used and important controls in the **normal working area** (forearm sweep ≈ 35 – 45 cm) and within the **maximum reach** (arm sweep ≈ 55 – 65 cm for 5th percentile female); occasionally used controls further away; heavy / large forces – hand-wheels / pedals at suitable height (≈ 90 – 110 cm for hand controls, elbow height; foot controls – ankle/knee height ahead).
- Use the **5th percentile reach** to decide the maximum distance and the 95th percentile for clearances; sitting reach differs from standing – consider clothing and seat position; leave space for gloves.
- Typical dimensions (check standards): push-button diameter 10 – 25 mm (finger), spacing ≥ 15 mm (10 mm for keyboards); toggle switch tip 3 – 25 mm, spacing 25 – 50 mm; knob diameter 10 – 100 mm, depth 12 – 25 mm; hand-wheel 175 – 525 mm; pedal 25 – 30 × 8 – 10 cm, travel 5 – 15 cm, force 20 – 90 N; emergency push-button ≥ 25 mm, red mushroom head.

## 9. General Design Checklist – Displays and Controls

| Displays – check | Controls – check |
|---|---|
| Is the right type of display used for the information (quantitative / qualitative / check reading)? | Is the right control type used for the task (discrete / continuous; hand / foot; force & precision)? |
| Is it visible and legible from the usual positions (distance, angle, light)? Character height ≥ 20 min arc? | Is it within reach of 5th percentile user, not obstructing the view and displays; correct height? |
| Are scales simple (1, 2, 5, 10), numerals upright, pointer touches marks, no parallax? | Are controls coded (shape, size, colour, label) & distinguishable; spacing sufficient for gloves? |
| Is the display grouped by function / sequence / importance; critical items in central 30° zone? | Are control–display relations compatible (stereotypes, Warrick, C/D ratio, same layout)? |

| Displays – check | Controls – check |
|---|---|
| Are warnings coded (red / amber / green), redundant (visual + auditory) & not too many? | Is resistance suitable (feedback, no accidental activation, guards for emergency / dangerous)? |
| Is glare, reflection, vibration, noise, darkness considered? Night lighting available? | Is the movement direction as expected (clockwise = increase)? feedback (click / light) provided? |
| Is the information load low (no clutter), consistent with standards and user training? | Can errors be reversed (undo / recover)? Is there an emergency stop & reset sequence? |
| Is maintenance (cleaning, replacement of lamps) easy; failure of display detectable? | Is the control robust, easy to clean, usable with PPE; tested by users? |

## 10. Standards for Ergonomics in Engineering Design

- Standards give **minimum requirements, dimensions, procedures and test methods** so that systems are safe and usable. They are developed by ISO, IEC, CEN (EN), BIS (India), ANSI / HFES, military (MIL-STD) and national bodies; they are revised periodically – always refer to the latest edition.

| Standard | Scope (ergonomics) |
|---|---|
| ISO 6385 | Ergonomic principles in the design of work systems (general principles) |
| ISO 9241 series (e.g. 9241-210, -11) | Ergonomics of human–system interaction: usability, human-centred design, display, keyboard, dialogue |
| ISO 7250 | Basic human body measurements for technological design (anthropometry) |
| ISO 14738 / EN 547 | Anthropometric requirements for the design of workstations at machinery (body dimensions) |
| ISO 9355 / EN 894 | Ergonomic requirements for the design of displays and control actuators (parts 1–3) |
| ISO 11064 | Ergonomic design of control centres (layout, workstations, displays, environment) |
| ISO 7731 / ISO 11428 | Auditory danger signals / visual danger signals (design, detectability) |
| IEC 60073 / IEC 60447 | Colours of indicator lights and push-buttons; actuating principles (coding) |
| ISO 11228 (parts 1–3) | Manual handling – lifting, carrying, pushing / pulling, repetitive tasks |
| ISO 7730, ISO 2631, ISO 5349 | Thermal comfort (PMV / PPD); whole-body vibration; hand-arm vibration |
| ISO 15534 / ISO 11226 | Dimensions for openings (access); evaluation of static working postures |
| ANSI / HFES 100, 200; MIL-STD-1472 | Human factors engineering of computer workstations & software; human engineering design criteria for military systems |

| Standard | Scope (ergonomics) |
|---|---|
| BIS (India): IS 3646 etc. | Indian standards, e.g. interior illumination (IS 3646) & others; Factories Act norms for noise / lighting / ventilation |
| OSHA / NIOSH guidelines | Workplace safety; lifting equation; noise & heat limits; ergonomic programmes |

Note: In answers, quote the principle of the standard (e.g. "ISO 9355 – coding & arrangement of displays & controls") rather than clause numbers.

## 11. Quick Revision & Probable Questions

### Memory hooks

- Cycle: Display → Sense → Perceive → Decide → Act → Control → Machine → (feedback).
- Visual displays: moving pointer / moving scale / digital; qualitative – quantitative – check-reading – representational.
- Visual: complex, permanent, quiet; Auditory: simple, urgent, moving person, dark; Tactile: eyes & ears busy.
- $H = D \times \theta / 3438$; $\theta$ = 20 – 22 minutes of arc; line of sight 15° below horizontal; C/D ratio optimum.
- Red = danger; Amber = caution; Green = normal; Blue = mandatory; White = neutral.

1. Explain the man–machine interaction cycle with a neat sketch. What are man–machine interfaces?
2. Discuss the factors that control the choice of a display. Compare visual, auditory and tactile displays.
3. Explain moving pointer, moving scale and digital displays with merits & demerits. How is a scale designed?
4. What are check-reading and representational displays? Explain indicators and colour coding of signal lights.
5. Explain the factors affecting effectiveness of displays.
6. Classify controls; explain coding of controls and their integration with displays (compatibility, C/D ratio).
7. Explain design guidelines for displays & controls: viewing distance, illumination, angle of view, reach. Give a general checklist.

8. Name the important standards for ergonomics in engineering design and displays & controls.
9. Numerical: find character height for $D = 700$ mm and $\theta = 20$ min.